

Article

Machine Learning-Based Cyber Fraud Detection: A Comparative Study of Resampling Methods for Imbalanced Credit Card Data

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Abstract

The prevalence of online transactions and extensive adoption of credit card payments have contributed to the escalation of credit card cyber fraud in modern society. These trends are propelled by technological advancements, which provide fraudulent actors with more opportunities. Fraudsters exploit victims' financial vulnerabilities by obtaining illegal access to sensitive credit card information through deceptive means, such as phishing, fraudulent phone calls, and fraudulent SMS messages. This study predicts and detects potential instances of cyber fraud in credit card transactions by employing Machine Learning (ML) techniques, including Decision Tree (DT); Random Forest (RF); Logistic Regression (LR); Support Vector Machine (SVM); K-Nearest Neighbors (KNN); XGBoost; CatBoost; and sampling techniques such as Tomek Link, Synthetic Minority oversampling technique (SMOTE), Edited Nearest Neighbor (ENN), Tomek+ENN, and SMOTE+ENN. To determine the performance of the algorithms in terms of accuracy, precision, recall, F1 score, and ROC-AUC for credit card cyber fraud detection, we conducted a comparative analysis of the extant ML techniques.

Keywords: credit card cyber fraud; fraud detection; artificial intelligence; machine learning; resampling techniques

1. Introduction

Credit card usage has experienced an enormous increase in recent times, owing to the constant advancement of technology. Consequently, this surge in usage has led to a progressive increase in the occurrence of cyber fraud associated with credit cards. Credit cards are the prevailing mode of payment in the modern business environments, encompassing both minor- and large-scale industries. The extensive integration of credit card cyber fraud has rendered it a widespread concern in numerous industries. The banking industry has experienced substantial transformation and benefited substantially from technological advancements. Nevertheless, the issue of credit card cyber fraud has been expedited owing to the fast rate of cyber adoption.

The banking industry faces significant difficulties owing to credit card cyber fraud, which causes financial losses amounting to billions of dollars annually [1,2]. In recent years, the acceptability of credit cards and other online payments has increased, and consequently, credit card cyber fraud has increased [3]. This makes cyber fraud detection an essential industry. Traditional approaches to address this problem are insufficient [4,5].



Academic Editor: Luis Javier García Villalba

Received: 22 November 2025

Revised: 24 December 2025

Accepted: 8 January 2026

Published: 14 January 2026

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The algorithms for the conventional method have been formulated by experts in the cyber fraud domain. In addition, a proactive approach is required to counteract cyber fraud. All sectors have endeavoured to implement Machine Learning (ML) based solutions owing to their widespread adoption, rapidity, and efficacy [6]. It has been established that the ML approach is among the most efficient approaches in this area [7]. The implementation of ML techniques has the potential to enhance the accuracy of monitoring system anomaly detection and threat reduction [8]. ML techniques facilitate extensive consumer behaviour surveillance and individual profiling, allowing for the tracking of their actions and the provision of actionable insights to aid in the mitigation of risks and cyber fraud [9]. To improve the data processing and analysis phases, credit card cyber fraud detection has implemented ML methodologies [10]. Despite the application of ML algorithms and data mining, efforts to mitigate cyber fraud have been ineffective, revealing challenges, such as false positives and negatives. False negatives cause financial losses for both cardholders and institutions, whereas false positives cause inconvenience and weaken the trust among cardholders. These difficulties continue to exist because of imbalanced data, adversarial attacks, concept dispersion, and model interpretability issues. It is imperative to conduct comprehensive assessments of ML algorithms using measures such as the precision, accuracy, recall, F1 score, and ROC-AUC. Additionally, implementing techniques, such as resampling, is crucial for boosting fraud-detection systems. Our study aims to avoid fraudulent attempts at credit cards in response to this imperative. Multiple ML algorithms were used to detect fraudulent activities. This study conducted an exhaustive evaluation of the performance of various ML techniques in this domain to identify the most efficient technique for enhancing the security of credit card transactions. In addition, this study presents the implementation of resampling methods as a strategy to counterbalance the data and prevent the inference of erroneous conclusions from the results. Through the application of cutting-edge ML techniques, our objective is to develop a resilient model that can accurately detect and combat instances of credit card cyber fraud. A comparative analysis of the various algorithms will yield significant insights into their individual capabilities and drawbacks, thereby enabling the determination of the most appropriate algorithms. The primary objective of this study is to examine the current deficiencies in techniques for detecting credit card cyber fraud and lay the groundwork for establishing more robust and efficient cyber fraud detection systems.

- **Cyber fraud in credit card**

Cyber Fraud is criminal conduct in which an individual in a position of authority fraudulently steals funds or exploits an organization's resources for personal advantage [11]. Illegal conduct has a substantial impact on the economy, legal systems, and social structure. Significant efforts have been made to combat fraudulent actions in the financial industry, resulting in the development of methods to detect cyber fraud. These strategies are included in the systems designed to prevent and detect cyber fraud, encompassing a range of processes to stop and uncover fraudulent acts. Credit card cyber fraud is categorized as online, where fraudsters use stolen credit card information, and offline, where personal details are obtained through skimming or phishing. The primary focus of this study was to explore online credit card cyber fraud. The growth of credit cards worldwide has surpassed 2.8 billion at present, resulting in an upsurge in fraudulent operations. Global credit card cyber fraud is anticipated to increase to \$43 billion by 2026, according to projections. The urgency for heightened security measures is underscored by recent statistics, which reveal a 46% surge in credit card cyber fraud attempts. Considering the dynamic nature of transaction environments, these trends emphasize the critical need for heightened security measures and increased awareness to combat the escalating issue of credit card cyber fraud [12]. The global credit card cyber fraud losses are shown in Figure 1.

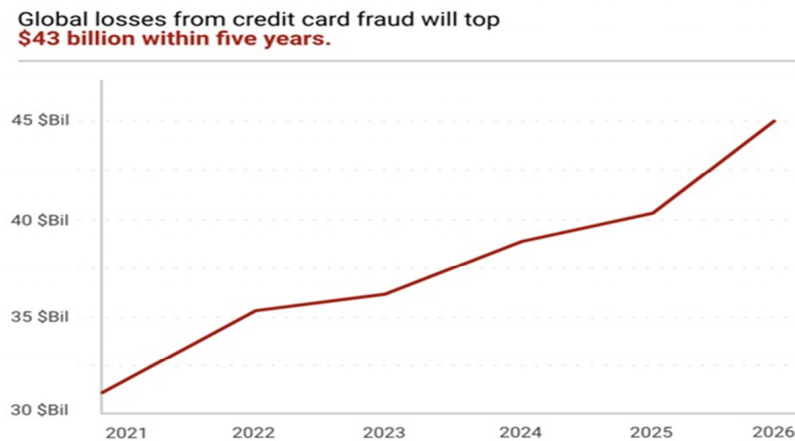


Figure 1. Global credit card cyber fraud losses reported across recent years.

- Challenges with cyber fraud detection systems

Cyber Fraud detection systems face major challenges, mostly related to imbalanced class distribution and large amounts of data. An imbalanced class distribution creates a challenge in properly detecting fraudulent transactions, because of the unequal representation of fraudulent and non-fraudulent cases. This imbalance may cause classification models to struggle to achieve consistent performance across many classes, possibly by completely ignoring minority classes [3]. The large amount and complexity of data in datasets makes detection methods more challenging. Data-reduction strategies are frequently used to address the issues posed by excessive dimensionality. Principal component analysis (PCA) is widely employed to reduce dimensionality, offering advantages such as faster calculations and improved data privacy through anonymization [13].

To detect credit card cyber fraud, we evaluated a variety of ML algorithms and sampling techniques, including Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), XGBoost, and CatBoost, with emphasis on precision, recall, and F1 score. Specifically, we considered the Tomek Link, Synthetic Minority oversampling technique (SMOTE), Edited Nearest Neighbor (ENN), combinations of SMOTE and ENN, and Tomek and ENN. Aligning resampling techniques with business objectives was emphasized. By conducting an exhaustive analysis of the algorithm performance metrics, our study facilitates the process of selecting models for cyber fraud detection. We emphasize the significance of maintaining a balance between false positives and false negatives using an exhaustive comparative analysis. On the other hand, the study conducted by [14] center on the difficulties associated with class imbalances in credit risk assessment. Although both studies discuss credit card cyber fraud, our study provides a comprehensive analysis of ML and practical recommendations for enhancing the accuracy of financial transaction detection. Ileberi et al. (2021) [15] focused on a specific ML framework for credit card cyber fraud detection, specifically addressing class imbalance using AdaBoost on a European credit card cyber fraud dataset. Alarfaj et al. (2022) [16] emphasized that the deep learning approach focuses on showcasing the superiority of deep learning models and exploring future avenues for improvement. Alamri and Ykhlef (2024) [17] introduced BCBSMOTE, a hybrid sampling method, to balance skewed credit card transaction datasets by using a random forest classifier for cyber fraud detection. Although previous research has focused on credit card cyber fraud detection, our study provides a more comprehensive analysis by examining different ML techniques and resampling strategies, thereby shedding light on the intricate details of cyber fraud detection.

The primary contribution of this study is the proposal of a machine learning-based cyber fraud detection framework tailored to extreme class imbalance. The proposed model integrates comprehensive data preprocessing, multiple resampling strategies, and a suite of classical machine learning classifiers within a leakage-safe and statistically supported evaluation protocol. By emphasizing precision–recall analysis, operating-point behaviour, and comparative stability, the framework provides practical guidance for selecting and deploying fraud detection models under real-world constraints.

The remainder of this paper is organized as follows. Section 2 presents the related work, and Section 3 presents the methodology used in this study and also introduces evaluation and reflection. Section 4 presents a comparative study and discussion, and Section 5 presents the conclusions, limitations, and future directions.

2. Related Works

ML which empowers computers to accomplish tasks without explicit programming, has great potential for anticipating risks and identifying abnormal behaviours in datasets, including credit card theft. ML's capacity of ML to predict and identify trends in large datasets makes it powerful. ML methods include supervised, unsupervised, semi-supervised, and reinforcement learning [18]. Table 1 summarizes the related work.

2.1. Supervised Techniques

Supervised learning trains ML models with input and output labels. The model extracts patterns from the labelled input and output data during training and uses them in subsequent evaluations. Supervised learning comprises two subcategories: classification and regression.

(A) Classification Techniques

ML classification techniques aim to predict the class label associated with a specific data point accurately. This is particularly relevant in cyber fraud detection, where the task involves forecasting whether a transaction is fraudulent or legitimate [19]. The SVM was applied to a maximum margin hyperplane to divide the data samples into two groups [20]. A new model involving a comparison between the SVM Classifier and tree-specific DT algorithm was conceptualized and implemented in a study by [21]. The findings suggest that the DT Classifier achieved a lower accuracy of 94.86% than the SVM Classifier, which achieved 98.59% accuracy. The four categories of kernel functions utilized by [22] are linear, radial, polynomial, and sigmoid, and this study employed by integrating the cuckoo search (CS) algorithm, genetic algorithm (GA), and particle swarm optimization (PSO) techniques. Experimentally, the linear kernel function was demonstrated to be the most effective. A radial basis function was employed to optimize the kernel function. The PSO-SVM exhibited superior overall efficacy compared to CS-SVM and GA-SVM. Nama et al. (2023) [23] introduced a new model. An SVM classifier is implemented using a multilayer perceptron learning (MLP) technique. The research findings indicated that the SVM and MLP techniques achieved an accuracy of 94.59% and 91.21%, respectively.

DT has garnered significant attention from researchers. In comparison with another classifier, Bandyopadhyay et al. (2021) [24] found that the DT algorithm outperformed the other with an accuracy of 99%. Choubey and Gautam (2020) [25] employed a variety of supervised algorithms including DT, RF, LR, naive Bayes (NB), and KNN. They found that the hybrid classifier DT with KNN performed more effectively than any individual classifier did. Hammed and Soyemi (2020) [26] described the application of a DT algorithm with a regression analysis as an enhancement. This outcome signifies an improvement in performance. The accuracy of this method was demonstrated by an 18.4% misclassification error rate, and the system validated every inserted incursion utilized for testing.

For classification, a Logistic Model Tree (LMT) was implemented in DT. Hussein et al. (2021) [27] implemented LMT for the classification and detection of cyber fraud. This outcome indicates that the utilization of the LMT algorithm for cyber fraud classification outperforms alternative methodologies. The LMT model achieved an accuracy of 82.08%.

One of the most powerful techniques in credit card cyber fraud detection is RF, which is a modern variation of DT. Amusan et al. (2021) [28] use RF to detect cyber fraud using skewed data. The findings revealed that RF achieved a maximum accuracy of 95.19% in comparison with LR, DT, and KNN. RF was implemented in conjunction with SVM, NB, and KNN as described by [29]. The results demonstrate that the RF algorithm outperforms the alternatives. The applications of LR, RF, and Catboost for the detection of cyber fraud were discussed in [30]. As demonstrated, RF with CatBoost yielded an excellent precision. RF produced the most precise results at 99.5%). Rai and Dwivedi (2020) [31] implemented an RF classifier-based method to detect cyber fraud in credit card systems. Comparison of existing classifiers LR and NB on multiple credit card system datasets. RF performed better than its competitors did. RF achieved an accuracy of 99.95%, whereas LR and NB achieved 91.16% and 89.35% accuracy, respectively. Ileberi et al. (2022) [32] proposed a new model using a genetic algorithm (GA), DT, RF ANN, LR, and NB. Results showed that GA-RF achieved an overall optimal accuracy of 99.98%.

According to Bayesian theory, Naive Bayes bases its decisions on maximum probability. The Bayesian probability approximates uncertain probabilities with known values. Analyzing uncertain claims requires reasoning and prior information. The conditional independence assumption implies that data features are independent [33]. NB and resilient scalability techniques for cyber fraud detection have been described by [34]. Fraud prediction is best with NB on a robust scale. NB robust scaling yielded 97.78% accuracy. Gupta et al. (2021) [35] found that NB outperformed RF, LR, and SVM with accuracy 80.4% and an area under the curve of 96.3%.

Neighborhood samples were utilized by KNN to determine the class designations. The KNN method performs the best [36]. A comparison was made between KNN and other ML techniques in [37]. The results demonstrate that the KNN model is the most accurate. The KNN algorithm generates optimal outcomes, including statistical measures. The f1-score was 82%, while the precision was 95%, and the recall was 72%.

(B) Regression Techniques

Alenzi and Aljehane (2020) [38] proposed a classifier constructed using the LR in a new system. The proposed LR-based classifier was compared with KNN and voting. With 97.2% accuracy, LR-based yields the best results. Itoo and Singh (2021) [39] compared the cyber fraud detection capabilities of LR, KNN, and NB. In contrast, LR surpassed both KNN and NB. The LR achieved an accuracy of 95%. Soh and Yusuf (2019) [40] proposed four cyber fraud detection methods for imbalanced data. The results suggest that both RF and KNN are overfitting. DT and LR were compared. The results indicated that DT made an error of 0.6%, whereas LR with stepwise division criteria was superior. Single, hybrid undersampling, and oversampling models were used by [41]. The analysis revealed that LR was the optimal algorithm. LR and neural networks (NN) are superior to DT in terms of the model performance.

(C) Ensemble Techniques

Bagging is a learning procedure that improves an estimator by creating several other estimators. The new ensemble model proposed by [42] incorporates the KNN, SVM, RF, bagging, and boosting classifiers. They have added undersampling and SMOTE to the ML algorithm. The results showed that the ensemble model reduced the false positives and negatives. The Boosting algorithms include XGBoost, AdaBoost, RUSBoost, Gradient Boosting Machine (GBM), and LightGBM. Faraj et al. (2021) [43] conducted a comparative analysis of ensemble methods to predict credit card cyber fraud. The results show that XGBoosting exhibits a superior performance. During stacking, a single model integrated the predictions of the contributing models. According to [44], a layered ensemble could be used to identify fraudulent transactions. The stacked ensemble achieved the highest performance of 0.78% after training with the sample dataset. Mienye and Sun (2023) [45] used deep-learning-based stacking ensembles to efficiently identify credit card cyber fraud. The meta-learner is an MLP, whereas the stacking ensemble uses LSTM and GRU neural networks as the basis learners. The proposed method demonstrated exceptional performance.

2.2. Unsupervised Techniques

The process of classifying comparable instances into identical clusters is called clustering. In contrast to the classification methods, clustering methods were considerably less prevalent in the reviewed literature. The K-means algorithm is a non-hierarchical technique used to aggregate data. Abdulsalami et al. (2019) [46] implemented a back-propagation neural network (BPNN) with k-means clustering. The outcome demonstrates that BPNN and K-means differ significantly in their ability to detect cyber fraudulent cyber credit card transactions. Comparatively, the BPNN model produced fewer false alarms and obtained a greater accuracy than the K-means model. The findings also indicated that the accuracy of BPNN was 93.1%, whereas the accuracy of K-means was 79.9%.

The isolated forest ensemble was operated without supervision. Point-based distance calculations and profiling performed on regular instances were absent. The Isolation Forest consisted of a collection of DTs. Jaiswal et al. (2021) [47] proposed a model that employs isolated forest and local outlier factor algorithms. The accuracy of the Local Outlier Factor was 97%, whereas that of the isolated forest was 76%. Oluwasanya and Braimah (2023) [48] introduced a model that combines LR and isolation forests. The findings derived from the evaluation of the dataset indicate that the isolation forest algorithm performed less effectively than the LR algorithm. Self-organizing map (SOM) learning was performed using unsupervised neural networks (NN). The SOM is suitable for constructing and analyzing consumer profiles to identify instances of cyber deception. Deb et al. (2021) [49] presented three unsupervised algorithms: SOM, K-distributed stochastic neighbor embedding (T-SNE), and K-means clustering using principal component analysis (PCA). The model achieved accuracy of 90%. The findings additionally demonstrate that k-means clustering in conjunction with PCA outperforms the basic k-means clustering. In addition, T-SNE is significantly superior to PCA, which is significantly affected by the outliers.

2.3. Semi-Supervised Techniques

An approach that integrates supervised and unsupervised learning methodologies. Prior to commencing the predictions, the supervised learning attribute examined the relationships within the representation, whereas the unsupervised learning attribute was used to ascertain the optimal data illustration. Semi-supervised methods have been proven to be highly advantageous in situations involving asymmetrical data acquisition. Dzakiyullah et al. (2021) [50] proposed a model that integrated autoencoders with semi-supervised learning. By employing an autoencoder, the proposed model subsequently trained a basic

linear classifier to classify the collected data into their respective categories. The T-SNE was implemented to visually distinguish between legitimate and illegitimate transactions. With 0.98% classification accuracy, credit card cyber fraud is readily identifiable, which is a useful result.

Based on the reviewed literature, this study focuses on supervised machine learning methods as the primary empirical approach. While unsupervised and semi-supervised techniques are valuable in scenarios with limited or unlabeled data, the availability of a fully labeled benchmark dataset enables a controlled and reproducible evaluation of supervised classifiers. Moreover, ensemble-based supervised models such as Random Forest, XGBoost, and CatBoost have consistently demonstrated strong performance in prior fraud detection studies, particularly under class imbalance. Accordingly, these models were selected as logical candidates for empirical evaluation, alongside baseline classifiers (DT, LR, KNN, and SVM), to enable a comprehensive and systematic comparison under identical preprocessing, resampling, and evaluation conditions.

2.4. Deep Learning (DL)

DL is a recent facet of ML that mimics the human brain to process diverse data, such as sights, sounds, and words. Unlike traditional ML, DL excels with large datasets by using unsupervised or semi-supervised feature learning. It plays a vital role in cybersecurity and effectively detects attacks effectively [51]. DL combines pattern recognition, text, and image classification and leverages algorithms such as CNN and LSTM. When applied to credit card cyber fraud detection, DL consistently outperforms the traditional methods [16]. Nevertheless, this study concentrates on ML methods and does not include DL techniques to detect cyber fraud.

Table 1. A summary of related work.

Reference	ML Techniques	Results	Dataset
[40]	LR, DT, KNN, RF	LR performance is better among others with 0.6% error rate.	Kaggle
[41]	LR, DT, KNN, RF, SVM, NB	LR performance is better than others with 96.99% accuracy. Precision 95%.	Kaggle
[46]	Kmean, BPNN	Results show that the accuracy of BPNN is 93.1%. Precision is 84%. Recall is 95%.	Kaggle
[25]	RF, DT, KNN, LR	DT+KNN Sensitivity is 85.63%. Precision 86.90%.	Kaggle
[26]	DT+ regression analysis	Accuracy is 81.06%.	UCI
[30]	LR, RF, Catboost	RF performs better with 99.95% accuracy. Precision is 91.95%. Recall is 79.2%.	Kaggle
[31]	RF, LR, NB	RF is better than its competitors with 99.95% accuracy.	Kaggle
[29]	RF, SVM, NB, KNN	RF performs better with 98.23% accuracy. Sensitivity is 97.02% Specificity is 98.23%.	UCI
[44]	ANN, GBM, SMOTE, Stacking	Stacked Ensemble had the best average performance at 78%. Precision is 86%.	Kaggle
[37]	LR, KNN	KNN produced best result with 99%. Precision is 95%. Recall 73%.	Kaggle
[38]	LR, KNN	LR is better with 97.2% accuracy. Sensitivity is 97%.	Kaggle
[22]	SVM, CS, GA, PSO	PSO-SVM accuracy is 98.05%.	Kaggle
[24]	DT	DT accuracy is 99%.	Kaggle
[27]	Logistic model tree	LMT is better with Accuracy of 82.08%. Precision is 80.40%. Recall is 82.10%.	UCI
[28]	RF, KNN, LR, DT	RF performs better with accuracy of 95.19%. Precision is 0.9794%. Recall is 0.9226%.	Kaggle
[34]	NB + robust scalars	Accuracy is 97.78%. Precision is 99.79%. Recall is 97.87%.	Kaggle
[35]	NB, LR, SVM, RF	NB is the best with accuracy 80.4%. Area under the curve is 96.3%.	Kaggle
[39]	LR, NB, KNN	LR is produced best result with 95% accuracy	Kaggle
[43]	KNN, DT, ANN, LR, XGBoost	XGBoosting performs better with accuracy is 68.76. Precision is 71.42. Recall is 61.58.	UCI
[47]	Isolation Forest, Local Outlier	Accuracy of Local Outlier Factor is 97%. Isolation Forest is 76%	Kaggle
[49]	K-means, PCA, T-SNE, SOM	This model achieved accuracy of 90%.	Statlog
[50]	AE, T-SNE, Gaussian	Precision is 98%. Recall is 98%.	Kaggle
[15]	RF, ANN, NB, LR	GA-RF achieved optimal accuracy of 99.98%. Precision is 99.92%. Recall is 99.82%.	Kaggle

Table 1. *Cont.*

Reference	ML Techniques	Results	Dataset
[21]	SVM, DT	SVM accuracy is 98.59%. Precision 100. Recall 96%.	Kaggle
[23]	SVM, MLP	SVM accuracy is 94.59%. MLP accuracy is 91.21%.	Kaggle
[45]	LSTM, GRU, MLP, Stacking	DL ensemble with the SMOTE-ENN achieved a sensitivity 1.00. Specificity 99.70%.	Kaggle
[48]	Isolation Forest, LR	LR is better with accuracy 78%. Precision is 95%. Recall 56%.	Kaggle
[42]	KNN, SVM, RF, SMOTE	The ensemble model reduced FP and FN with 99.96 accuracy.	Kaggle
[52]	LR, RF, XGBoost, SMOTE	XGBoost showed high AUC (0.97). RF achieved AUC of 0.98 with strong F1-score performance.	Kaggle
[53]	LR, RF, GBM, MLP, SMOTE	RF: achieved 99.999% accuracy. GBM performed 99.98% accuracy	Kaggle

3. Materials and Methods

This section outlines the credit card dataset utilized in the study and offers a thorough explanation of the different algorithms and techniques employed in developing the proposed credit card cyber fraud detection approach.

3.1. Dataset

To assess the effectiveness of the suggested ML models, a widely recognized dataset was chosen for both training and testing. This dataset is available at <https://www.kaggle.com/mlg-ulb/creditcardfraud> (accessed on 20 January 2020). This dataset remains widely used in recent credit card fraud detection studies (e.g., [3,54]), supporting its continued relevance as a benchmarking dataset under extreme class imbalance. The dataset consists of client transactions at a European bank in 2013. The real-world dataset consists of 284,807 credit card transactions. The dataset consists of 99.828% legitimate transactions and 0.172% fraudulent transactions. A summary of the research datasets used in this study is provided in Table 2.

Table 2. A summary of the dataset utilized in this study.

Source	https://www.kaggle.com/mlg-ulb/creditcardfraud
Type	Real-world/Numerical
Size	284,807
Category	Extremely imbalanced
Feature	Most of the data has been anonymized.
Feature size	31
Features	Amount: The amount of the transaction.
	Time: Cumulative duration, measured in seconds, between the initial transaction and under examination.
	Class variable: (0 equals not fraud, 1 equals fraud)
	[V1 . . . V28]: The output obtained from conducting PCA for dimensionality reduction on the original raw data is utilized to safeguard the identity of users and secure critical characteristics.

3.2. Programming Language

Python 3.11.4, an easy-to-use, interpreted, object-oriented, and high-level programming language, has shown a trend in recent years. Python is the top ML language. This approach provides powerful ML tools and libraries. Numpy, which performs numerical operations for multidimensional arrays and linear algebra, was employed in the proposed system. Pandas allow for fast and flexible data analysis and manipulation. Python Scikit-learn was used for statistical and ML models.

3.3. Classification Techniques

The dataset was employed to train multiple ML models using RF, DT, LR, KNN, SVM, XGBoost, and CatBoost algorithms. After training the model, its performance was evaluated, followed by the generation of a comparative visual representation that illustrates the distinctions between each of the algorithms.

(A) Random forest (RF)

Multiple DTs are used in the RF method to detect credit card cyber fraud. Each tree uses a random collection of data and features. Combining the tree forecasts yielded the final projection. RF handles credit card transaction data with large dimensions and complex feature dependencies. Figure 2 shows the RF algorithm.

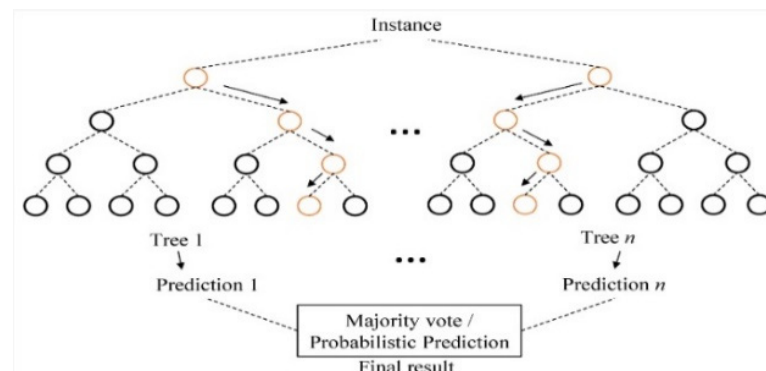


Figure 2. Architecture of the Random Forest (RF) classification algorithm.

The initial RF generation and prediction are the main stages of RF. Information gathering is required to determine whether to divide two nodes or root nodes. Entropy and Gini impurities were the primary criteria for acquiring information. Gini impurity is defined as the probability that the input data will be incorrectly classified using the classification provided.

(B) Decision trees (DT)

DTs are hierarchical structures that are used for classification or regression. The structure comprises nodes that symbolize feature tests interconnected via branches, resulting in a multitude of outcomes. When detecting credit card cyber fraud, DTs construct a tree-like structure by iteratively partitioning data according to feature values. Decision-making is rendered transparent by utilizing if-then-else logic at the nodes to classify the transactions in these trees. DTs are susceptible to overfitting, which restricts generalization and renders them excessively customized to the training data despite their intuitive appearance. Effective cyber fraud detection requires a delicate equilibrium between the level of complexity and depth of the tree.

(C) Logistic regression (LR)

LR is a statistical method that models a binary dependent variable by using a logistic function. LR ascertains the likelihood of a binary response by applying a functional methodology and an assortment of characteristics. This process uses a non-linear sigmoid function to ascertain the parameters that yield the optimal fit. The sigmoid function (σ) and its associated input (x) are formulated as follows [33]. Through the individual multiplication of each element, the vector z representing the input data and the optimal coefficients w are generated. The sum of these quantities yields a numerical value that establishes the classification score for the target class. When the sigmoid value is below 0.5, it is zero; otherwise, it is assigned a value of one. LR derives its nomenclature from the foundation of linear regression, a widely employed ML technique designed for regression tasks. Figure 3 shows a sigmoidal function graph.

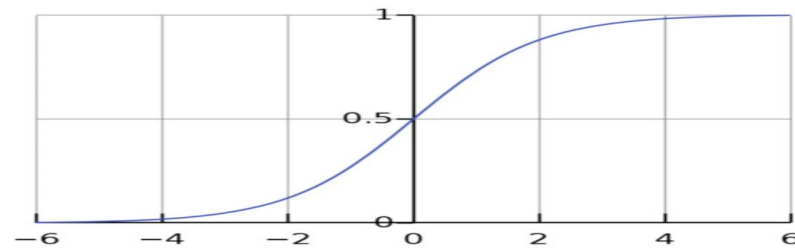


Figure 3. Sigmoid function graph.

(D) K-nearest neighbors (KNN)

The KNN classifier is an instance-based learning approach that utilizes Euclidean, Minkowski, and Manhattan distances among other similarity measures. The Minkowski distance is most effectively applied to categorical variables, whereas Euclidean and Manhattan distances are more suitable for continuous variables [33]. The Euclidean distance between the current input and an alternative input data point is computed for every data point in the dataset. Following the ascending arrangement of the computed Euclidean distances, k items were selected according to the item that was near the input data in terms of distance. The input point is assigned to the classification by the classifier that occurs most frequently among k data points.

(E) Support vector machine (SVM)

SVMs are classification methods that seek to identify the most effective hyperplane for separating the data elements into their respective classes. In credit card cyber fraud detection, the SVM identifies the best decision boundary to distinguish genuine from fraudulent transactions. It determines the support vectors and points closest to the decision border. SVM demonstrates high efficacy in datasets with numerous dimensions, making it well-suited for credit card data with multiple attributes. Notably, SVMs can discern decision boundaries in complex, high-dimensional regions, making them suitable for scenarios in which data segregation is not immediately apparent. Additionally, SVMs can handle non-linear data patterns using kernel functions, enhancing their ability to capture intricate linkages in credit card transactions. Figure 4 shows the SVM graph.

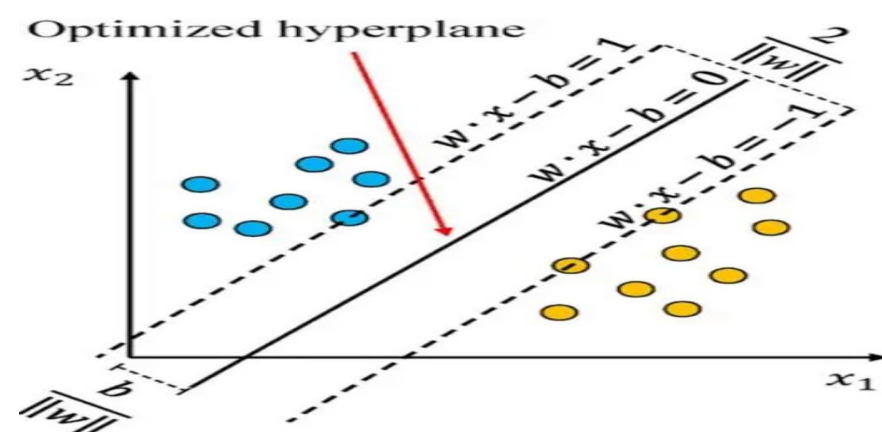


Figure 4. Support Vector Machine (SVM) decision boundary illustration.

(F) XGBOOST

XGBoost is a widely used technique that utilizes an additive learning approach with second-order approximation. It involves a first-order derivative known as the gradient, and a second-order derivative known as the Hessian of the loss function. XGBoost has emerged as a prominent algorithm in practical ML applications. XGBoost is a boosted gradient

technique that is based on decision trees. XGBoost is preferred over alternative gradient boosting machines (GBMs) owing to its memory efficiency, fast execution speed, and effective model performance. XGBoost uses parallel computation to build trees by utilizing all the available CPUs during the training process. XGBoost enhances computational performance and speed by utilizing the “max depth” parameter for tree pruning in the backward direction rather than the typical stopping criteria.

(G) CATBOOST

CatBoost is a strong contender among the classifiers for highly imbalanced data. CatBoost is a gradient boosting method designed to effectively handle categorical and ordered features while also addressing overfitting with Bayesian estimators. CatBoost does not require substantial data training compared to other ML models and can be effectively used with many types and forms of data. CatBoost includes CPU and GPU implementation. When employing ensembles of comparable sizes, the GPU iteration of the GBDT outperformed other prominent open-source GPU implementations, including XGBoost and LightGBM, and facilitated faster training. CatBoost employs a more efficient technique that mitigates overfitting and enables utilization of the entire dataset for training.

3.4. Resampling Methodology

Most predictive models perform inferiorly when the distribution of the classes is imbalanced. Consequently, data preprocessing must be executed prior to supplying the model with data as input. When confronted with a class imbalance issue, data preprocessing is executed using a data-level methodology known as the resampling approach. Resampling can be broadly classified into three distinct approaches: undersampling, oversampling, and hybrid sampling. The undersampling method reduces the majority class to achieve balance in the dataset. It is optimal to implement when the dataset is enormous, and reducing the preponderance of samples can significantly improve performance and alleviate storage issues. The oversampling technique is diametrically opposed to the undersampling technique. The replication of observations from the minority class equalizes the proportions of majority and minority samples. Finally, a hybrid approach combines the undersampling and oversampling techniques to achieve rebalancing.

(A) Tomek link removals

The Tomek-link is composed of two instances of separate classes that are near each other. A pair (E1, E2) is considered a Tomek Link when two samples E1 and E2 from distinct classes are given; there is no sample E3 in which the distance between E1 and E3 is less than the distance between E1 and E2 or E2 and E3. The removal of Tomek links can be considered as an undersampling method involving the elimination of the plurality sample contained within the link.

(B) Synthetic minority oversampling technique (SMOTE)

SMOTE is a widely employed method to address class imbalances in datasets. The objective is to generate synthetic cases for the minority class by employing interpolation techniques among the nearest minority examples, instead of oversampling with replacement. Consequently, this mitigates overfitting. The selection of the nearest neighbors for minority cases is conducted randomly, with the number of neighbors chosen depending on the desired level of oversampling.

(C) The edited nearest neighbor (ENN)

ENN, the prevalent undersampling algorithm, eliminates samples by determining whether the classes of multiple samples and k-nearest neighbors are identical. The ENN method involves undersampling to address the class imbalance in classification tasks. ENN identifies misclassified samples in the majority class using their nearest neighbors, which have a different class label from the majority class. This eliminates incorrectly categorized samples and efficiently reduces the size of dominant classes.

3.5. Leakage-Safe Evaluation Pipeline

To ensure a fair evaluation and prevent data leakage, all preprocessing and resampling operations were performed strictly within the training data only. The complete experimental pipeline followed a leakage-safe procedure, summarized as follows:

1. The dataset was first split into training and testing sets using an 80:20 ratio. The test set was held out and never used during preprocessing, resampling, or model training.
2. For experiments involving k-fold cross-validation ($k = 5$), all preprocessing steps were applied independently within each training fold.
3. Feature scaling using StandardScaler was fitted exclusively on the training data (or training fold) and then applied to the corresponding validation or test data.
4. Resampling techniques, including Tomek Links, SMOTE, ENN, SMOTE+ENN, and Tomek+ENN, were applied only to the training data after scaling and were never applied to the test set.
5. Machine learning models were trained using the pre-processed training data and evaluated on the untouched test data.

This pipeline ensures that no information from the test set leaks into the training process. The adopted workflow follows best practices for imbalanced learning and is consistent with leakage-safe implementations such as those supported by the imbalanced-learn pipeline framework. At no stage was the test set involved in feature scaling, resampling, model selection, or hyperparameter tuning, ensuring a strictly leakage-safe evaluation protocol.

3.6. Evaluation and Reflection

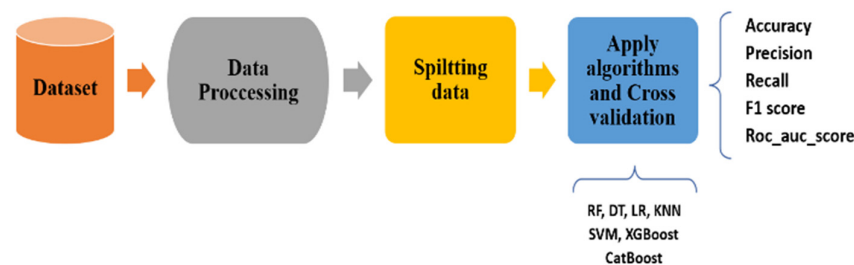
In this study, dataset imbalance is handled using k-fold cross-validation ($k = 5$) applied exclusively to the training data for model selection and hyperparameter tuning. Final performance results are reported on a held-out test set that is never exposed during training, preprocessing, or resampling. It utilizes metrics such as confusion matrix, accuracy, recall, precision, F1-score, and AUC-ROC. True positive (TP), true negative (TN), false positive (FP), and false negative (FN) values are crucial for evaluating a classifier's ability to accurately identify instances of credit card cyber fraud. The confusion matrix ensures a balance between selectivity and specificity and minimizes Type I and Type II errors. Despite the accuracy limitations of imbalanced datasets, this study emphasizes the importance of recall, precision, and F1-score. AUC-ROC was highlighted for its effectiveness in unbalanced data scenarios. However, given the extreme class imbalance in credit card fraud detection, precision–recall area under the curve (PR-AUC), also referred to as Average Precision (AP), is additionally reported and emphasized, as it provides a more informative evaluation of minority-class performance and positive predictive value than ROC-AUC. Table 3 presents the performance indicators.

Table 3. Performance indicators and evaluation metrics used for credit card fraud detection.

Metrics	Description	Equation	Range
Accuracy	Assess the number of TPs	$A = \frac{TN + TP}{TN + FN + TP + FP}$	[0–1]
Recall	The ratio of TP to a TP and FN	$R = \frac{TP}{TP + FN}$	[0–1]
Precision	The ratio of TP to a TP and FP	$P = \frac{TP}{TP + FP}$	[0–1]
F1-Score	Combines precision and recall	$F_1 = 2 \frac{P \times R}{P + R}$	[0–1]
AUC	The area between two points bounded by the function and the x axis.	$AUC = \int_a^b f(x)dx$	[0–1]
PR-AUC (AP)	Area under the precision–recall curve, emphasizing minority-class prediction performance	$AP = \sum_n (R_n - R_{n-1}) \cdot P_n$	[0–1]

4. Method

The proposed model is illustrated in Figure 5. To facilitate the development of the cyber fraud detection model, a dataset containing credit card transaction information is obtained and loaded into the analysis environment.

**Figure 5.** Proposed model for detecting credit card cyber fraud.

The format of the dataset file was Comma-Separated Values (CSV). The CSV file can be read using the Panda module in Python. Exploring data insights is achieved through diverse visualization techniques, marking the primary step in the data analysis process. In ML, the pre-processing stage is significant because it improves data quality, facilitating the extraction of relevant knowledge from extensive datasets more efficiently. Moreover, it ensures that the data are suitable for modeling purposes. This represents the starting point for the creation and implementation of the ML models. Data preprocessing involves cleaning and structuring the data and addressing issues such as partial, inaccurate, missing values, or inconsistencies. After progressing through this phase, the target attribute was examined, revealing a substantial imbalance between the proportions of Classes 1 and 2. This significant class imbalance could potentially lead to biased data. Imbalanced data refers to categorization difficulties characterized by disparate quantities of cases across distinct classes. Imbalanced data are prevalent in the generic datasets. Cyber Fraud detection systems often encounter large amounts of highly imbalanced data, where most credit card transactions are genuine, but only a small fraction is fraudulent. To assess the presence of data imbalance in relation to the goal variable, namely the class, we proceed with an examination of the data. Figure 6 shows the percentage of fraudulent and non-fraudulent transactions. The imbalanced distribution of classes is shown in the bar chart in Figure 7.

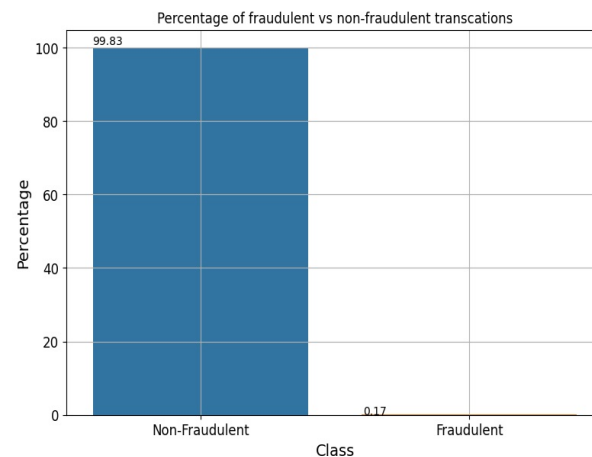


Figure 6. Percentage distribution of fraudulent versus non-fraudulent transactions in the dataset.

Because of the dataset's profound imbalance, with legitimate transactions accounting for 99.827% and cyber fraud transactions for only 0.173%, training ML models using the raw data are unlikely to yield reliable outcomes, despite potentially high evaluation metrics. Before executing the ML algorithms, it was therefore necessary to preprocess the data. This is because different models impose distinct requirements on the predictors, and the quality of the training data directly affects the prediction output. The primary objectives of data preprocessing are to cleanse and transform the data into a form that reduces bias, addresses missing or inconsistent values, and enhances variability for effective learning.

The dataset exhibits extreme class imbalance, with fraudulent transactions accounting for approximately 0.173% of all observations, corresponding to an imbalance ratio of roughly 1:577 (fraudulent to legitimate transactions). In the machine learning literature, datasets with minority class proportions below 1% are commonly regarded as extremely imbalanced. This level of imbalance is substantially more severe than that observed in many standard benchmark datasets and poses significant challenges for model training and evaluation.

The dataset comprises numerical values derived from a Principal Component Analysis (PCA) process. Nevertheless, the original feature descriptions have not been released owing to confidentiality constraints. In total, thirty features are provided, of which twenty-eight were obtained through PCA. PCA is a widely used dimensionality-reduction technique that transforms high-dimensional data into a lower-dimensional representation by identifying a smaller set of components that retain the most informative variance of the original features. The only attributes that have not undergone conversion into principal components are Amount and Time.

The Time attribute represents the elapsed seconds from the first recorded transaction rather than a true timestamp. As such, it does not encode calendar effects, customer behavioural cycles, or genuine temporal ordering. Consequently, its inclusion may introduce spurious correlations without improving model generalization. For this reason, the Time feature was excluded to focus the analysis on transaction-level behavioural attributes, which is consistent with common practice in prior credit card cyber fraud detection studies using this dataset.

Preprocessing tasks were carried out using the Python data manipulation library pandas and the scikit-learn machine learning framework. Following the importation of the credit card dataset into the Python environment, a comprehensive data-cleansing procedure was applied. Data cleaning typically involves two main activities: the removal of null or missing values and the handling of outliers, which are observations that deviate

substantially from the majority of the data. In this dataset, no null or missing values were detected.

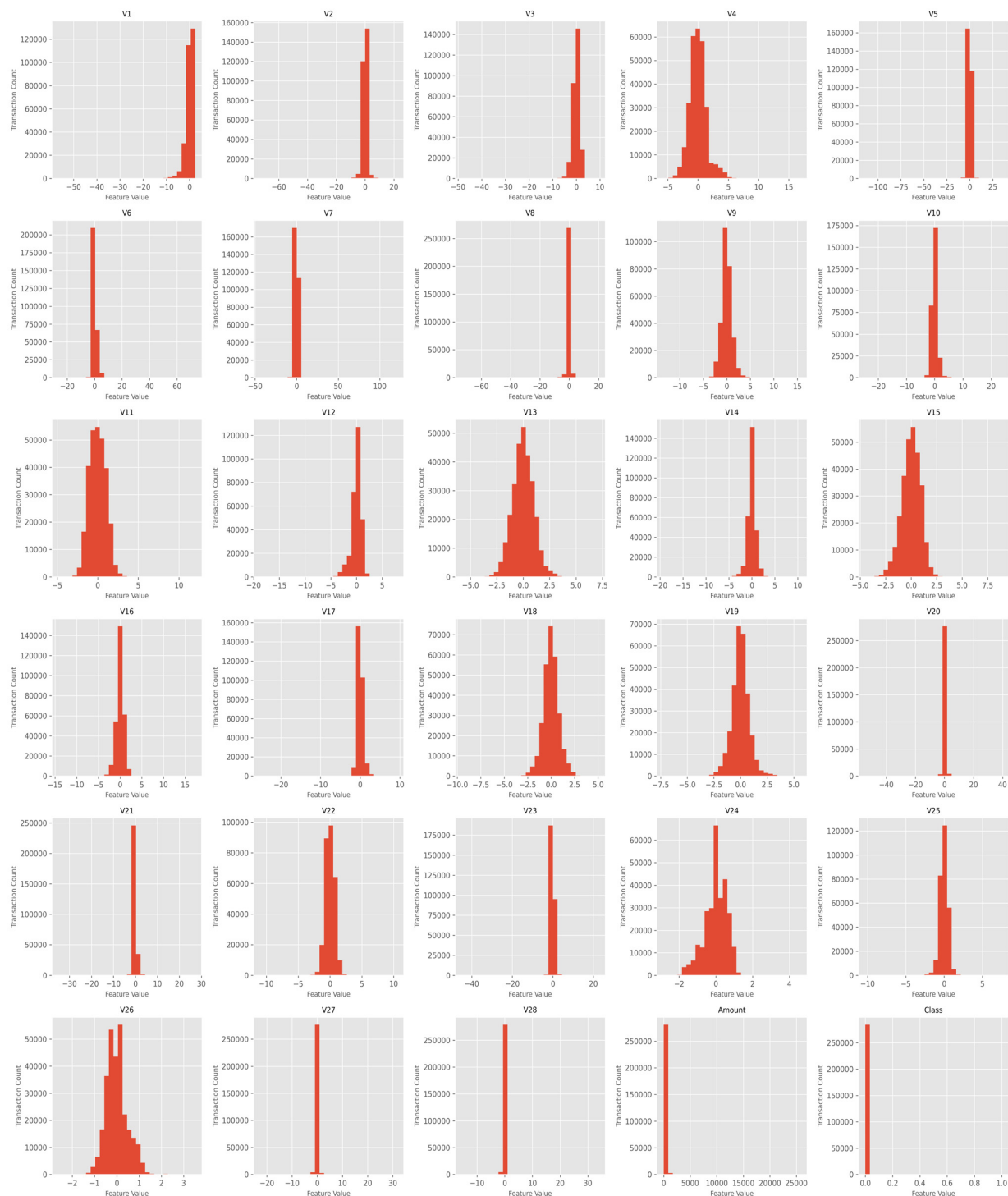


Figure 7. Feature-wise histograms illustrating the distribution of feature values (x -axis) and transaction counts (y -axis) across the dataset.

Feature selection in credit card cyber fraud detection involves the careful selection of a subset of relevant attributes from the initial feature set. The objective is to retain the most discriminative and informative features while eliminating redundant or less useful ones, thereby enhancing model performance, reducing overfitting, and improving interpretability. A heatmap and correlation matrix are presented in Figure 8 to support

this analysis. The correlation matrix assists in identifying features that may be candidates for removal. Although all features exhibit some degree of association with the Class label, and the dataset contains thirty-one attributes, certain features may be more beneficial to the models than others. Accordingly, during the preprocessing stage, superfluous characteristics were removed. Given that the most informative features had already been extracted through PCA, the Time attribute was excluded from the final feature set, as it was not deemed to provide additional predictive value.

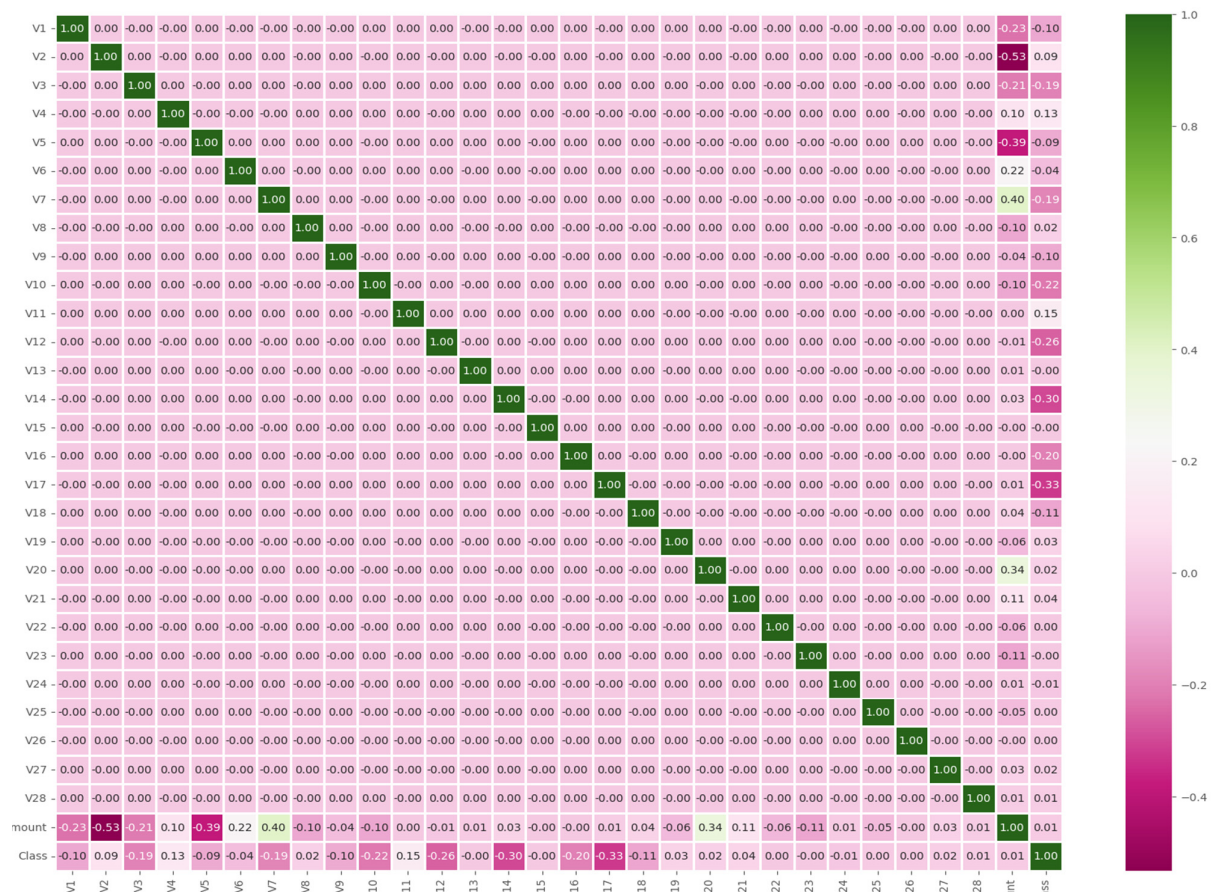


Figure 8. Correlation matrix showing relationships between input features and the class label.

Preliminary data preprocessing consisted of separating features from the objective variable ('Class') and dividing the data into training and testing sets at an 80–20 ratio. After removing missing values and standardizing features using StandardScaler, resulting in X_{train} and X_{test} , the imbalanced dataset undergoes division for ML algorithms (RF, DT, LR, KNN, SVM, XGBoost, and CatBoost). Evaluation metrics like accuracy, precision, recall, and AUC assess algorithmic performance. Cross-validation with five folds on the training data yielded scores, followed by predicting labels for the test set and calculating the classification metrics (precision, recall, F1 score, and AUC). A heatmap and ROC curve provided insights into classification accuracy and model effectiveness, contributing to the evaluation of the research paper. This study investigated a range of algorithms, employing GridSearchCV to optimize them prior to assessment using critical metrics such as Accuracy, Precision, Recall, F1 Score, and ROC-AUC. The evaluative framework goes beyond AUC diagrams, addressing AUC limitations in severely imbalanced datasets and offering a comprehensive understanding of each algorithm's efficacy in cyber fraud detection.

Initially, we assessed the performance of the ML algorithms in the absence of any sampling technique. The results of this evaluation are presented in Table 4, which provides

a comparative analysis of the algorithms. Figure 9 shows the performance of the ML algorithms before applying resampling techniques. Figure 10 shows the Confusion Matrix for ML techniques. All results reported in Table 4 are obtained from evaluation on the held-out test set, while cross-validation is used only during training for hyperparameter optimization. To reflect performance variability and model stability, cross-validation results obtained on the training data are summarized using mean $\pm$ standard deviation across five folds, while the primary conclusions are drawn from the held-out test set results reported in Tables 4 and 5.

Table 4. Algorithm performance before applying resampling techniques.

Algorithm	Accuracy	Precision	Recall	F1 score	ROC-AUC	PR-AUC (Average Precision)
RF	0.9995	0.9625	0.7857	0.8652	0.8928	0.8682
DT	0.9994	0.9195	0.8163	0.8649	0.9081	0.8199
LR	0.9992	0.8462	0.5612	0.6748	0.7805	0.7567
KNN	0.9995	0.9500	0.7755	0.8539	0.8877	0.8080
SVM	0.9993	0.9683	0.6224	0.7578	0.8112	0.8527
XGBoost	0.9996	0.9620	0.7755	0.8588	0.8877	0.8778
CatBoost	0.9995	0.9875	0.8061	0.8876	0.9031	0.8890

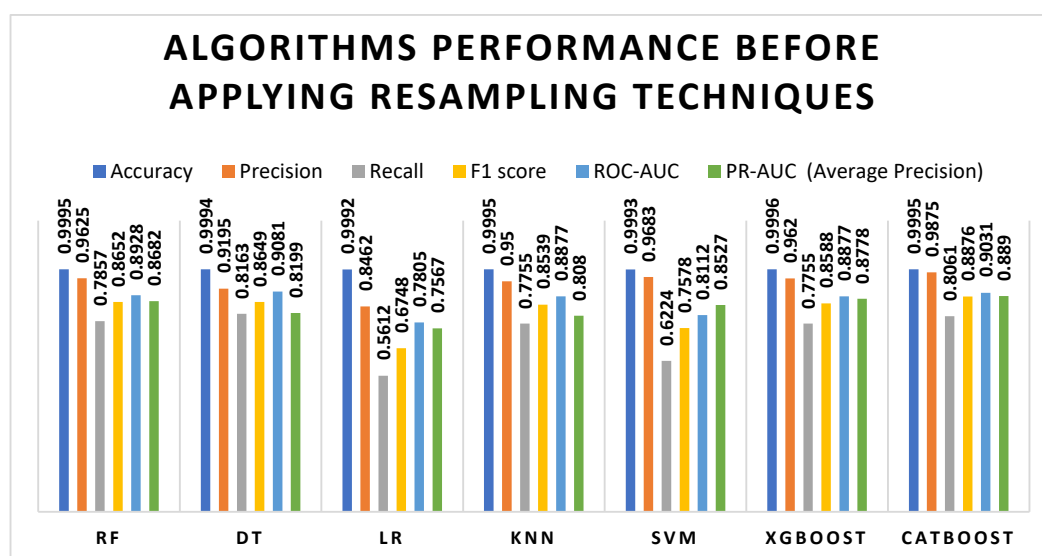


Figure 9. Performance of machine learning algorithms before applying resampling techniques.

Table 5. Cross-validation stability on the training set (5-fold CV), reported as mean $\pm$ SD of PR-AUC.

Technique	PR-AUC (5-Fold CV Mean $\pm$ std)
Random Forest	0.8431 $\pm$ 0.0342
CatBoost	0.8416 $\pm$ 0.0386
XGBoost	0.8471 $\pm$ 0.0415

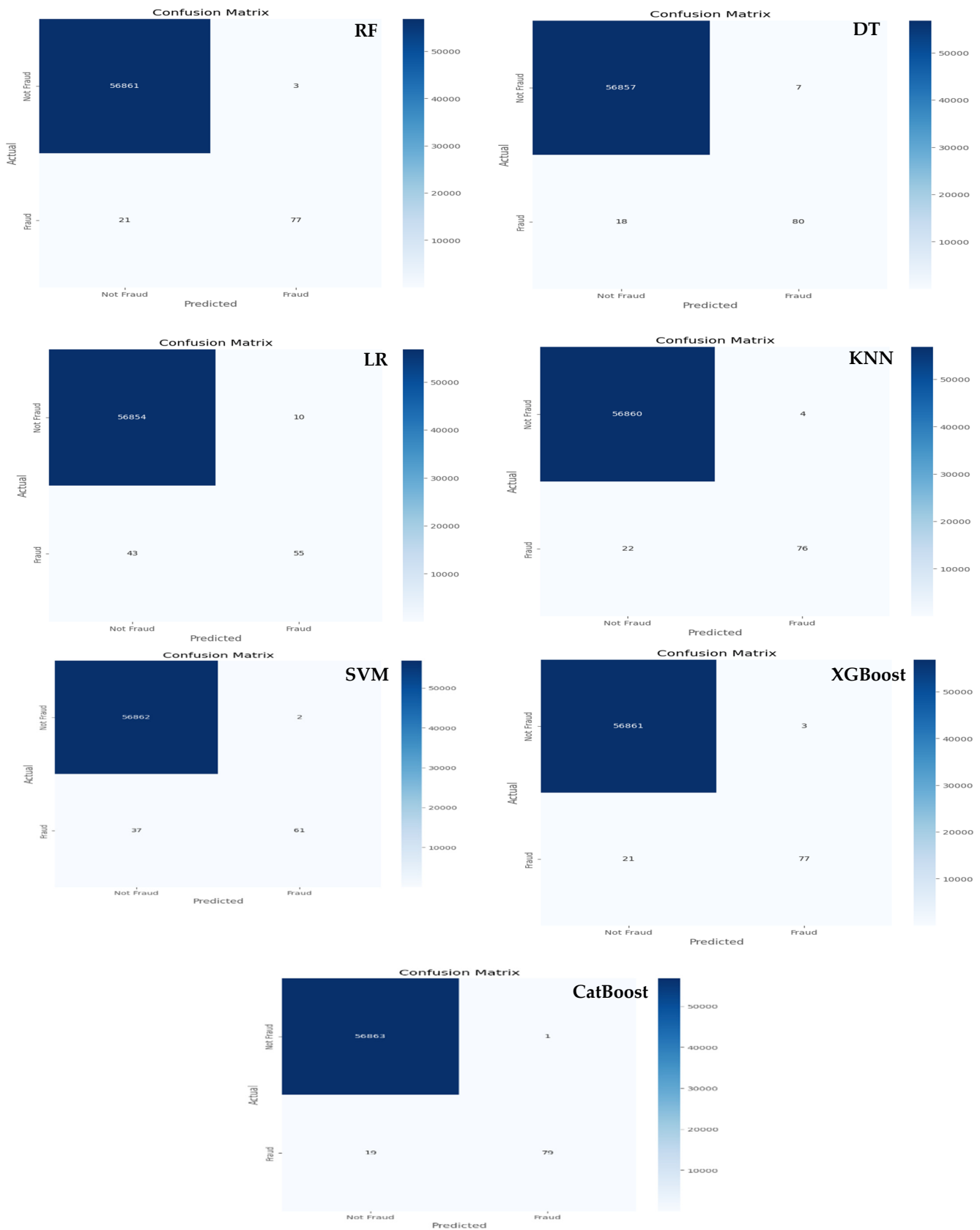


Figure 10. Confusion matrices for ML algorithms before applying resampling techniques.

These results indicate comparable stability across folds for the top-performing models, while the primary comparisons and conclusions remain based on the held-out test set results reported in Tables 4 and 5.

- Comparison between ML techniques before applying resampling

The classifier in the RF structure is configured with the critical parameters that govern its operation. The RandomForestClassifier was initialized with a forest consisting of 300 estimators (trees), with a maximum depth of 20 levels per tree. To split an internal node, a minimum of five samples was stipulated (`min_samples_split` = 5). Similarly, to split a leaf node, a minimum of two samples were required. This configuration balances the model complexity and prevents overfitting, ensuring that each tree is sufficiently deep and can make meaningful splits while avoiding excessive complexity. RF performed well in correctly identifying positive cases, evident from the high precision of 96.25%. However, the recall was relatively low at 78.57%, indicating a potential challenge in capturing all instances of the minority class. The F1 score (86.52%) strikes a balance between precision and recall, and the ROC-AUC (89.28%) confirmed its effectiveness in overall classification.

In the DT structure, we configured the classifier with specific parameters to control its behaviour. The decision tree classifier was set with a maximum depth of 40 levels (`max_depth` = 40), ensuring that the tree could capture complex relationships within the data. Setting the minimum number of samples necessary to divide an internal node to 8 and the minimum number of samples needed to split a leaf node to 10. “Entropy” is specified as the criterion for node separation, indicating that information gain is utilized to quantify node quality. Like RF, the DT outperformed in precision (91.95%) and recall (81.63%). The F1 score (86.49%) and ROC-AUC (90.81%) signify the model’s ability to make informed decisions based on input features. Therefore, the simplicity and interpretability of this method are advantageous.

The LR classifier was created using specific parameters including a regularization term (L2 penalty) and regularization strength ($C = 0.1$). This aims to prevent overfitting and to improve the generalization of the model. LR showed a trade-off between precision (84.62%) and recall (56.12%). This imbalance resulted in a moderate F1 score (67.48%) and ROC-AUC (78.05%). The model may struggle to identify instances of the minority class, thereby indicating the potential need for feature engineering or parameter tuning.

The KNN was configured with parameters to customize its behaviour. The choice of `n_neighbors` = 3 determines the number of neighbors considered during the prediction, providing a balance between the local and global patterns. Setting `weights` = ‘uniform’ means that all neighbors have equal influence on the prediction, whereas $p = 2$ specifies the Euclidean distance metric for the distance calculation. The `algorithm` = ‘auto’ parameter adapts the algorithm for the nearest neighbor search based on the training data. These adjustments allow for fine-tuning of the KNN model to better suit the characteristics of the dataset and the desired predictive performance. KNN excels with balanced precision (95.00%) and recall (77.55%). The model effectively captured patterns in the data, leading to a robust F1 score (85.39%) and ROC-AUC (88.77%). Its simplicity and effectiveness render it a competitive choice.

An SVM was constructed using tailored parameters. Parameter C regulates the intensity of regularization, which assists in striking a balance between accurately classifying the training points and attaining a seamless decision boundary. The ‘rbf’ kernel, denoting a kernel established on Radial Basis Function, is selected due to its appropriateness for non-linear decision boundaries. The gamma parameter, set to ‘scale,’ determines the kernel coefficient for the ‘rbf’ kernel. Finally, `probability` = true enables the model to provide the probability estimates. SVM achieved high precision (96.83%) but faced challenges in recall (62.24%). The model’s performance was reflected by a moderate F1 score (75.78%) and ROC-AUC (81.12%). Addressing recall imbalance can enhance overall effectiveness.

The XGBoost model training algorithm employed standard hyperparameters to achieve effectiveness. The algorithm is composed of 100 boosting cycles (trees), oper-

ates at a learning rate of 0.1, has a maximum tree depth of three, and a minimal child weight of one. The gamma parameter controls the minimum loss reduction required for the further partitioning of the leaf node. Additionally, subsample and colsample_bytree define the ratios of the training instances and features sampled during the tree construction, respectively. The objective was set to 'binary: logistic,' which indicates a binary-classification task. When appropriately tuned, these parameters contribute to the robustness and predictive performance of the classifier. XGBoost model maintained a balanced precision (96.20%) and recall (77.55%), resulting in an impressive F1 score (85.88%) and ROC-AUC (88.77%). The ensemble-based approach proved effective in capturing the complex relationships within the data.

The CatBoost with hyperparameters 'depth' set to 1, 'learning_rate' at 0.8, 'n_estimators' of 300, and 'l2_leaf_reg' set to 7, demonstrated robust performance in the classification task. A low 'depth' value suggests a shallow tree structure, and a high 'learning_rate' typically aids faster convergence during training. The 'n_estimators' parameter specifies the number of boosting rounds, and 'l2_leaf_reg' controls the regularization strength. CatBoost stands out as the most robust model, achieving excellent in precision (98.75%) and recall (80.61%). The F1 score (88.76%) and ROC-AUC (90.31%) underscored their ability to handle imbalanced datasets effectively. The strong performance of the model positions it as the preferred choice for this classification task, particularly in scenarios with Imbalanced classes.

In summary, RF, DT, XGBoost, and CatBoost demonstrate remarkable performance in accurately classifying instances. These models strike a good balance between precision and recall, which is crucial for tasks that involve imbalanced datasets. LR struggled to balance precision and recall. KNN is simple and effective, making it a competitive choice. SVM exhibits high precision but faces challenges in terms of recall. Overall, CatBoost is the most robust option, particularly for imbalanced datasets. CatBoost is the most promising algorithm in terms of precision, recall, and overall performance. This demonstrates the superior capability to minimize false positives and effectively capture instances of cyber fraud. A thorough examination of these algorithms yields information that can be used to determine the optimal model for detecting cyber fraud involving credit cards. In addition to ROC-AUC, PR-AUC (Average Precision) was reported to better reflect minority-class detection performance under extreme class imbalance, with CatBoost and Random Forest achieving the highest PR-AUC values, confirming their superior ability to maintain high precision across varying recall levels. To statistically support this observation, a paired McNemar's test was conducted on the held-out test set at a fixed decision threshold of 0.5, comparing CatBoost and Random Forest. The results indicate that the observed performance difference is statistically significant ($p < 0.001$), providing additional support for the distinct predictive behaviour of the two models under extreme class imbalance.

- Operating-Point Analysis

To provide practical insight into model behaviour under severe class imbalance, an operating-point analysis was conducted at a fixed decision threshold of 0.5. At this threshold, precision, recall, and confusion-matrix counts (TP, FP, TN, FN) are reported on the held-out test set. This analysis highlights the trade-off between false positives and false negatives and complements threshold-independent metrics such as PR-AUC.

Further experimentation was conducted using the same set of ML techniques on newly generated datasets resulting from the application of various sampling techniques, namely, Tomek, SMOTE, ENN, SMOTE+ENN, and Tomek+ENN. The selected models were then trained on the modified datasets to obtain diverse analytical performances under identical training and testing set-splitting ratios.

- Comparison between ML techniques after applying resampling

After conducting a thorough evaluation of the effectiveness of ML algorithms without implementing sampling techniques, we introduced resampling techniques to further investigate and improve the performance of the algorithms. These resampling techniques include Tomek, SMOTE, ENN, SMOTE + ENN, and Tomek+ENN. This expanded exploration allowed us to scrutinize the algorithms under various conditions, shedding light on their robustness and adaptability in the context of imbalanced datasets.

The application of the Tomek, SMOTE, ENN, SMOTE + ENN, and Tomek+ENN techniques serves specific purposes in the context of handling imbalanced datasets in ML. Tomek is often applied to address the issue of overlapping instances between the minority and majority classes. Tomek links are pairs of extremely close instances (one from a majority class and one from a minority class). Removing these instances can improve the separation between classes. Figure 11 shows the performance of the ML algorithms after applying the tokek link. To correct the class imbalance, SMOTE is frequently employed to generate synthetic samples from minority classes. This contributes to balancing the class distribution and may improve the capacity of the model to extrapolate to instances belonging to the minority classes. Figure 12 shows the performance of the ML algorithms after applying SMOTE. The ENN aims to eliminate noisy examples by removing instances from the majority class that are misclassified by their KNN. It focuses on cleaning the dataset by removing potentially misleading majority class instances. Figure 13 shows the performance of the ML algorithms after applying ENN.

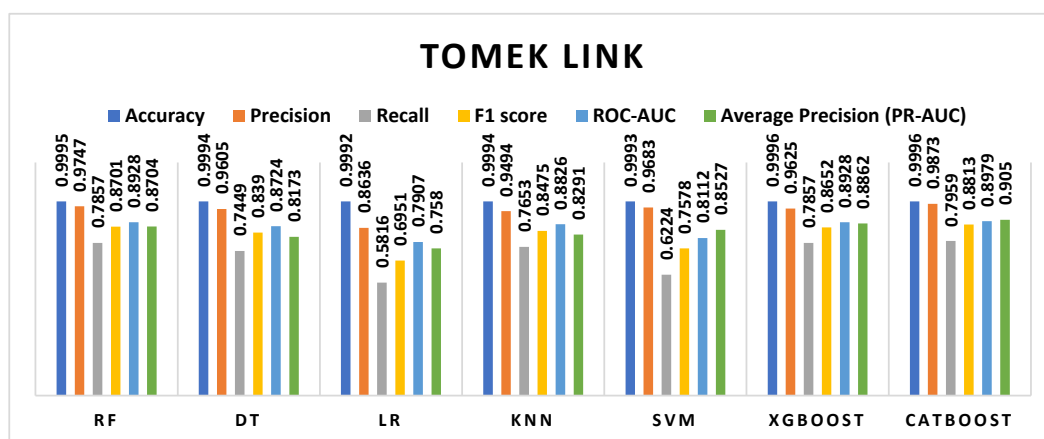


Figure 11. Performance of machine learning algorithms after applying Tomek Links resampling.

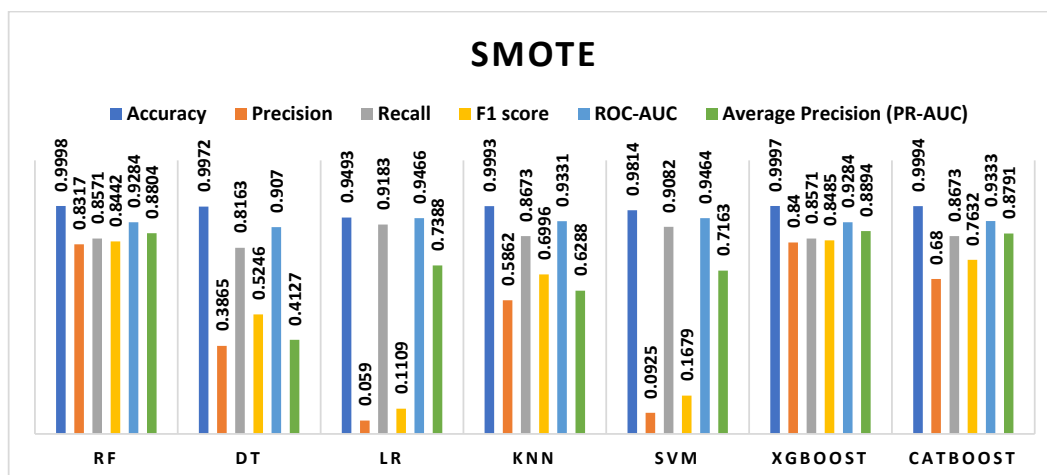


Figure 12. Performance of machine learning algorithms after applying SMOTE resampling.

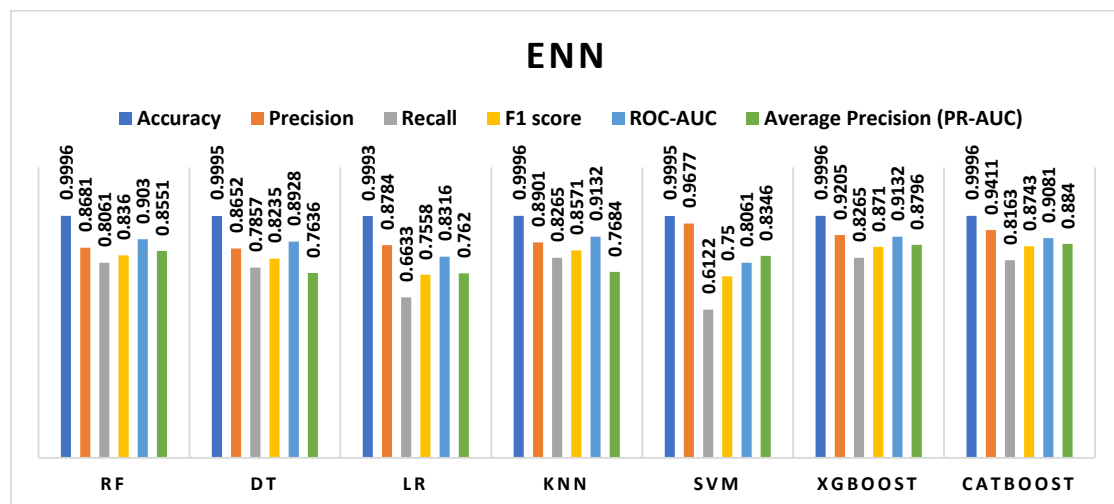


Figure 13. Performance of machine learning algorithms after applying ENN resampling.

Combining SMOTE with ENN involves first applying SMOTE to oversample the minority class and then using ENN to clean the dataset by removing noisy instances. This combination is useful when oversampling and cleaning are required. Figure 14 shows the performance of the ML algorithms after applying SMOTE + ENN. Similarly to SMOTE + ENN, this combination involves first using Tomek to remove instances that create Tomek links and then applying ENN for further cleaning. This is effective in scenarios in which both the Tomek links and noisy instances must be addressed. The outcomes derived from this evaluation are presented in Table 6, which illustrates a comparative analysis of the algorithms. Figure 15 shows the performance of the ML algorithms after applying TOMEK + ENN.

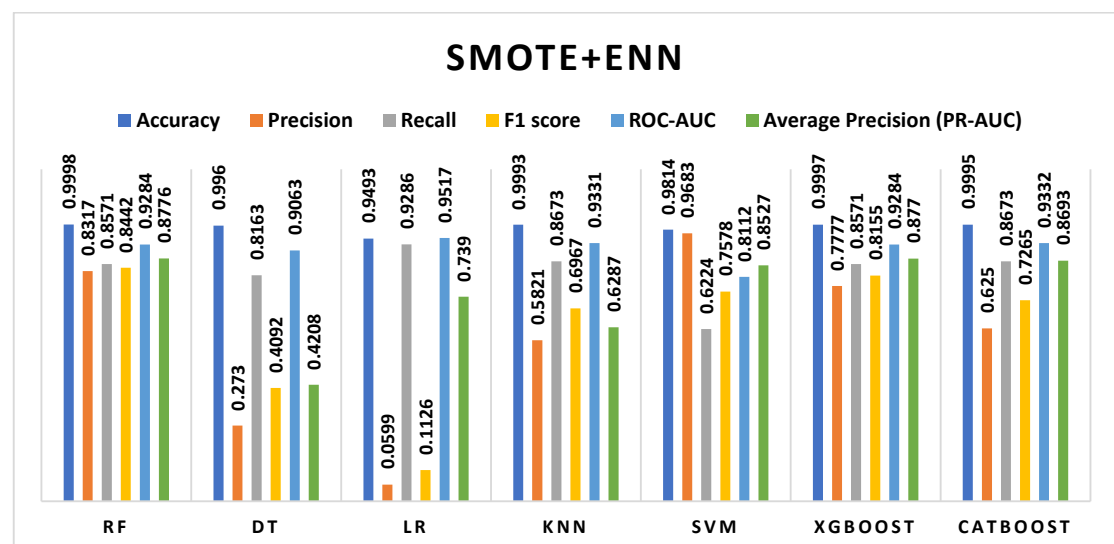


Figure 14. Performance of machine learning algorithms after applying SMOTE+ENN resampling.

Table 6. Algorithm performance after applying resampling techniques.

Resample	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Average Precision (PR-AUC)
Tomek	RF	0.9995	0.9747	0.7857	0.8701	0.8928	0.8704
	DT	0.9994	0.9605	0.7449	0.8390	0.8724	0.8173
	LR	0.9992	0.8636	0.5816	0.6951	0.7907	0.7580
	KNN	0.9994	0.9494	0.7653	0.8475	0.8826	0.8291
	SVM	0.9993	0.9683	0.6224	0.7578	0.8112	0.8527
	XGBoost	0.9996	0.9625	0.7857	0.8652	0.8928	0.8862
	CatBoost	0.9996	0.9873	0.7959	0.8813	0.8979	0.9050
Resample	Model	Accuracy	Precision	Recall	F1 score	ROC-AUC	Average Precision (PR-AUC)
SMOTE	RF	0.9998	0.8317	0.8571	0.8442	0.9284	0.8804
	DT	0.9972	0.3865	0.8163	0.5246	0.9070	0.4127
	LR	0.9493	0.0590	0.9183	0.1109	0.9466	0.7388
	KNN	0.9993	0.5862	0.8673	0.6996	0.9331	0.6288
	SVM	0.9814	0.0925	0.9082	0.1679	0.9464	0.7163
	XGBoost	0.9997	0.8400	0.8571	0.8485	0.9284	0.8894
	CatBoost	0.9994	0.6800	0.8673	0.7632	0.9333	0.8791
Resample	Model	Accuracy	Precision	Recall	F1 score	ROC-AUC	Average Precision (PR-AUC)
ENN	RF	0.9996	0.8681	0.8061	0.8360	0.9030	0.8551
	DT	0.9995	0.8652	0.7857	0.8235	0.8928	0.7636
	LR	0.9993	0.8784	0.6633	0.7558	0.8316	0.7620
	KNN	0.9996	0.8901	0.8265	0.8571	0.9132	0.7684
	SVM	0.9995	0.9677	0.6122	0.7500	0.8061	0.8346
	XGBoost	0.9996	0.9205	0.8265	0.8710	0.9132	0.8796
	CatBoost	0.9996	0.9411	0.8163	0.8743	0.9081	0.8840
Resample	Model	Accuracy	Precision	Recall	F1 score	ROC-AUC	Average Precision (PR-AUC)
SMOTE+ENN	RF	0.9998	0.8317	0.8571	0.8442	0.9284	0.8776
	DT	0.9960	0.2730	0.8163	0.4092	0.9063	0.4208
	LR	0.9493	0.0599	0.9286	0.1126	0.9517	0.7390
	KNN	0.9993	0.5821	0.8673	0.6967	0.9331	0.6287
	SVM	0.9814	0.9683	0.6224	0.7578	0.8112	0.8527
	XGBoost	0.9997	0.7777	0.8571	0.8155	0.9284	0.8770
	CatBoost	0.9995	0.6250	0.8673	0.7265	0.9332	0.8693
Resample	Model	Accuracy	Precision	Recall	F1 score	ROC-AUC	Average Precision (PR-AUC)
Tomek+ENN	RF	0.9996	0.8587	0.8061	0.8316	0.9029	0.8569
	DT	0.9995	0.8652	0.7857	0.8235	0.8928	0.7754
	LR	0.9993	0.8767	0.6530	0.7485	0.8265	0.7636
	KNN	0.9996	0.8901	0.8265	0.8571	0.9132	0.7682
	SVM	0.9995	0.9130	0.6429	0.7545	0.8214	0.7752
	XGBoost	0.9997	0.9205	0.8265	0.8710	0.9132	0.8805
	CatBoost	0.9996	0.9195	0.8163	0.8649	0.9081	0.8867

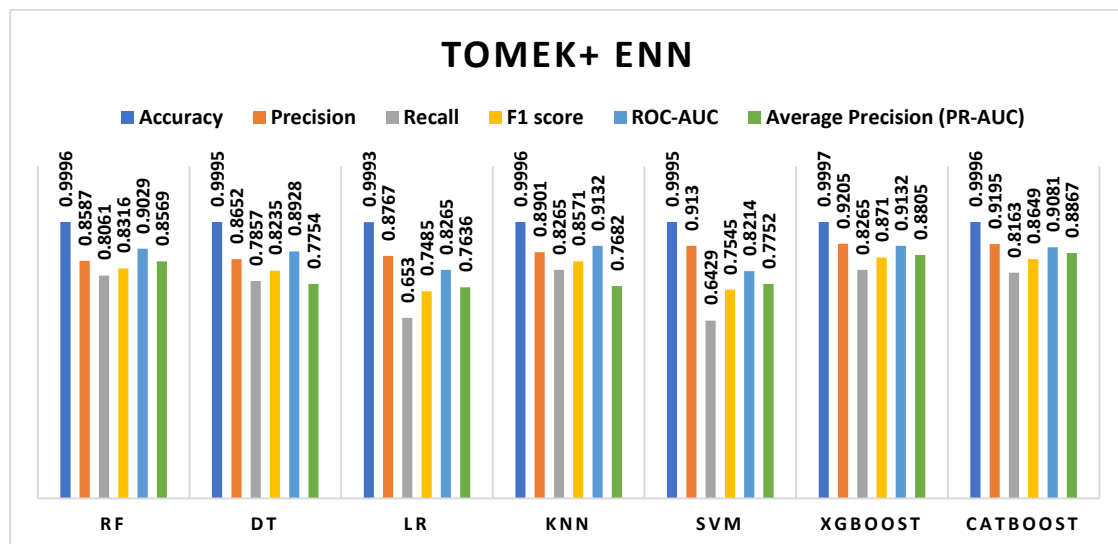


Figure 15. Performance of machine learning algorithms after applying Tomek+ENN resampling.

In addition to accuracy, ROC-AUC, and threshold-based metrics, precision–recall area under the curve (PR-AUC), also referred to as Average Precision, is reported to better reflect minority-class detection performance under extreme class imbalance. As shown in Table 6, ensemble-based classifiers, particularly CatBoost, XGBoost, and Random Forest, consistently achieve the highest PR-AUC values across most resampling strategies, indicating their ability to maintain high precision across a wide range of recall levels. By contrast, models such as Logistic Regression and Decision Tree exhibit lower PR-AUC scores under oversampling techniques (e.g., SMOTE), highlighting their increased sensitivity to false positives despite improved recall. These findings confirm that PR-AUC provides complementary and more informative insight than ROC-AUC alone in highly imbalanced credit card fraud detection tasks.

RF before resampling demonstrated strong overall performance with a precision of 96.25%, indicating its capability to accurately classify non-fraudulent transactions. However, there was room for improvement in recall (78.57%). The F1 score, providing a balanced assessment of precision and recall, stood at 86.52%. The ROC-AUC, measuring the model's discriminative ability, was robust at 89.28%. After Resampling, Tomek Links improved RF's precision to 97.47%, addressing the initial recall deficiency. The F1 score remained high, at 87.01%. The ROC-AUC remained strong at 89.28%. Additionally, the reduction in the dataset size owing to the Tomek Links might contribute to improved computational efficiency. SMOTE resulted in increased recall (85.71%) while sacrificing precision (83.17%). The F1 score decreased to 84.42%, and the ROC-AUC notably increased to 92.84%, demonstrating enhanced cyber fraud detection capability. ENN maintained high precision (86.81%) and improved both of recall (80.61%) and ROC-AUC at 90.30%. while decreased the F1 score to 83.60%. The combination of SMOTE+ENN achieved a notable balance, improving recall (85.71%) while sacrificing precision (83.17%). The F1 score decreased to 84.42% and the ROC-AUC increased to 92.84%. The synergistic effect of SMOTE for oversampling and ENN for undersampling contributes to a well-balanced cyber fraud detection model. The combined Tomek+ENN decreased precision (85.87%) and improved recall (80.61%), resulting in an F1 score (83.16%). The ROC-AUC -maintained robustness of 90.29%. The complementary nature of Tomek Links and ENN addresses the trade-offs between precision and recall, emphasizing the suitability of the model for various cyber fraud detection scenarios.

In summary, the choice of resampling technique significantly affects the RF performance. Tomek Links and Tomek+ENN demonstrated effectiveness in enhancing precision, with RF precision improving from 96.25% to 97.47% and 85.87%, respectively. This makes them suitable for tasks in which minimizing FP is crucial. On the other hand, SMOTE and SMOTE+ENN are beneficial for improving recall, as seen in the increase from 78.57% to 85.71% and 85.71%, respectively, making them valuable in scenarios where capturing more instances of cyber fraud is a priority.

DT before Resampling demonstrated robust performance. The precision, recall, F1 score, and ROC-AUC for DT were 91.95%, 81.63%, 86.49%, and 90.81%, respectively. This indicates a well-balanced ability to classify instances of cyber fraud. After Resampling, Tomek Links aimed to enhance the precision and recall balance. The results showed a slight increase in precision to 96.05%, but there was a noticeable decrease in recall of 74.49%. This trade-off resulted in a lower F1 score (83.90%) and ROC-AUC (87.24%). While the Tomek Links contributed to precision, they also led to a reduction in recall. The SMOTE led to a significant decrease in precision of 38.65%. Recall decreased to 81.63%, resulting in a lower F1 score (52.46%). However, the ROC-AUC remained relatively high at 90.70%. Synthetic samples from SMOTE enhanced the model's ability to capture instances of cyber fraud but at the expense of precision. ENN was effective in maintaining high precision (86.52%) while slightly sacrificing recall to 78.57%. This trade-off resulted in a balanced F1 score of 82.35% and a strong ROC-AUC of 89.28%. The ENN's focus on removing noise and redundant samples contributes to a refined balance between the precision and recall. The combination of SMOTE and ENN aims to achieve balance. However, there was a significant decrease in precision to 27.30% and a high recall of 81.63%, resulting in a lower F1 score (40.92%) and ROC-AUC (90.63%). The synergistic effect of SMOTE and ENN contributed to a well-balanced but slightly reduced precision-recall trade-off. The combination of Tomek+ENN resulted in a balanced precision of 86.52% and recall of 78.57%, contributing to a competitive F1 score (82.35%) and a robust ROC-AUC (89.28%). The complementary nature of Tomek Links and ENN addresses the trade-offs between precision and recall, making the model suitable for various cyber fraud detection scenarios.

In summary, resampling techniques induced trade-offs in the precision and recall of the DT model. Tomek Links and Tomek+ENN demonstrated effectiveness in maintaining or slightly improving precision, which is suitable for scenarios that prioritize the minimization of false positives. SMOTE and SMOTE + ENN were effective in enhancing recall, which is valuable in scenarios where capturing more instances of cyber fraud is a priority. ENN offers a balanced approach, and the choice of resampling technique depends on the specific objectives and desired trade-offs in precision and recall.

LR before resampling model initially demonstrated a decent overall performance with a precision of 84.62%, indicating its capability to accurately classify non-fraudulent transactions. However, there was room for improvement in recall (56.12%), suggesting the model might miss some instances of cyber fraud. The F1 score stood at 67.48%. The ROC-AUC, measuring the model's discriminative ability, was at a moderate level of 78.05%. LR after resampling and applying Tomek Links aimed to enhance precision and recall balance. However, the results showed no significant changes in precision (86.36%) or recall (58.16%). This resulted in the maintenance of the F1 score (69.51%) and ROC-AUC (79.07%). The SMOTE led to a significant decrease in precision to 05.98%, indicating a substantial increase in false positives. While recall increased significantly to 91.83%, the F1 score decreased to 11.09%. The ROC-AUC remained relatively high at 94.66%, showing an improved ability to capture instances of cyber fraud, but at the expense of precision. ENN was effective in maintaining a high precision of 87.84% and increased recall to 66.33%. This trade-off resulted in an improved F1 score of 75.58% and a strong ROC-AUC of 83.16%.

The combination of SMOTE and ENN aims to achieve balance. However, there was a significant decrease in precision to 05.99%) and increased recall to 92.86%, resulting in a lower F1 score (11.26%) and ROC-AUC (95.17%). Utilizing Tomek+ENN increased both precision to 87.67% and recall to 65.30%, contributing to a competitive F1 score 74.85% and a robust ROC-AUC 82.65%.

In summary, Tomek+ENN was effective in improving both the precision and recall. SMOTE and SMOTE+ENN, on the other hand, were effective in enhancing recall, but at the cost of precision. ENN offers a balanced approach, and the choice of resampling techniques depends on the specific objectives and desired trade-offs in precision and recall.

Before resampling, KNN demonstrated a strong overall performance. The precision, recall, F1 score, and ROC-AUC for KNN were 95.00%, 77.55%, 85.39%, and 88.77%, respectively. After Resampling, Tomek Links were applied to enhance the precision and recall balance. The results showed a slight decrease in precision to 94.94% and recall of 76.53%. This trade-off resulted in a lower, but still competitive, F1 score (84.75%) and ROC-AUC (88.26%). Tomek Links contributed to maintaining precision while addressing the initial recall deficiency. The SMOTE led to a significant decrease in precision of 58.62%. Recall increased to 86.73%, resulting in a decreased F1 score (69.96%). However, the ROC-AUC notably increased to 93.31%, showing an improved ability to capture instances of cyber fraud, but at the expense of precision. ENN was effective in maintaining a high precision of 89.01% and increased recall to 82.65%. This trade-off resulted in an increased F1 score of 85.71% and a strong ROC-AUC of 91.32%. The combination of SMOTE and ENN aims to achieve balance. However, there was a significant decrease in precision to 58.21%, an increase in recall of 86.73%, resulting in a lower F1 score 69.67% and ROC-AUC 93.31%. The combination of Tomek+ENN resulted in a balanced precision of 89.01% and improved recall to 82.65%, contributing to a competitive F1 score (85.71%) and a robust ROC-AUC (91.32%).

In summary, the resampling techniques induced trade-offs in the precision and recall of the KNN model. Tomek Links and Tomek+ENN demonstrated effectiveness in maintaining or slightly improving precision, which is suitable for scenarios that prioritize the minimization of false positives. SMOTE and SMOTE+ENN, on the other hand, were effective in enhancing recall, but at the cost of precision. ENN offers a balanced approach, and the choice of resampling technique depends on the specific objectives and desired trade-offs between precision and recall.

In SVM before resampling, precision, recall, F1 score, and ROC-AUC for SVM were 96.83%, 62.24%, 75.78%, and 81.12%, respectively. This indicates a well-balanced ability to classify instances of cyber fraud. After resampling, Tomek Links showed a slight increase in precision to 96.83% and recall to 62.24%. This trade-off resulted in a higher F1 score (75.78%) and ROC-AUC (81.12%). The SMOTE led to a significant decrease in precision of 09.25%. Recall increased to 90.82%, resulting in a decreased F1 score (16.79%). However, the ROC-AUC notably increased to 94.64%, showing an improved ability to capture instances of cyber fraud, but at the expense of precision. ENN was effective in maintaining a high precision of 96.77% and improved recall to 61.22%. This trade-off resulted in a balanced F1 score of 75.00% and a strong ROC-AUC of 80.61%. The combination of SMOTE and ENN aims to achieve balance. Precision remained high at 96.86%, and recall of 62.24%, resulting in F1 score (75.78%) and ROC-AUC (94.64%). The combination of Tomek+ENN resulted in a balanced precision of 91.30% and improved recall of 64.29%, contributing to a competitive F1 score (75.45%) and a robust ROC-AUC (81.12%).

In summary, Tomek Links and Tomek+ENN with SVM demonstrated effectiveness in maintaining or slightly improving both the precision and recall. SMOTE and SMOTE+ENN, on the other hand, were effective in enhancing recall, but at the cost of precision. ENN

offers a balanced approach, and the choice of resampling technique depends on the specific objectives and desired trade-offs between precision and recall.

The XGBoost Before resampling demonstrated strong overall performance; the precision, recall, F1 score, and ROC-AUC for XGBoost were 96.20%, 77.55%, 85.88%, and 88.77%, respectively. This indicates a well-balanced ability to classify instances of cyber fraud. After resampling, applying Tomek Links resulted in slight increase in both precision to 96.25% and recall to 78.57%. This trade-off resulted in a higher F1 score (86.52%) and ROC-AUC (89.28%). The SMOTE led to an increase in recall (85.71%) while sacrificing precision (84.00%). The F1 score decreased to 84.85%, and the ROC-AUC notably increased to 92.84%, showing an improved ability to capture instances of cyber fraud. ENN was effective in maintaining high precision (92.05%) and improved both recall to 82.65 and F1 score to 87.10%, and a strong ROC-AUC of 91.32%. The combination of SMOTE and ENN aims to achieve balance. There was a decrease in precision to 81.73%, an increase in recall to 86.73%, resulting in a slightly lower F1 score (84.16%) and a high ROC-AUC (93.35%). The combination of Tomek+ENN resulted in maintained high precision (92.05%) and recall (82.65%), contributing to a balanced F1 score (87.10%) and a robust ROC-AUC (91.32%).

In summary, the resampling techniques induced trade-offs in the precision and recall of the XGBoost model. Tomek Links and Tomek+ENN demonstrated effectiveness in maintaining or slightly improving precision. SMOTE and SMOTE+ENN were effective in enhancing recall, but at the cost of precision. ENN offers a balanced approach, and the choice of resampling technique depends on the specific objectives and desired trade-offs between precision and recall.

The CatBoost Before resampling demonstrated strong overall performance; precision, recall, F1 score, and ROC-AUC for CatBoost were 98.75%, 80.61%, 88.76%, and 90.31%, respectively. This indicates a well-balanced ability to classify instances of cyber fraud with a high precision and reasonable recall. After resampling, Tomek Links resulted in a slight decrease in precision to 98.37% and a modest decrease in recall to 79.59%. This trade-off resulted in a slightly lower F1 score (88.13%) and AUC score (89.79%). The SMOTE led to a decrease in precision to 68.00%, while increasing recall to 86.73%. The F1 score decreased to 76.32% and the ROC-AUC remained relatively high at 93.33%. Synthetic samples from SMOTE aimed to improve the model's ability to capture instances of cyber fraud but resulted in a trade-off with precision. ENN was effective in maintaining high precision (94.11%) and recall 81.63%. This trade-off resulted in a balanced F1 score of 87.43% and a strong ROC-AUC of 90.81%. The combination of SMOTE+ENN decreased precision to 62.50% and increased recall to 86.73%, resulting in a slightly lower F1 score (72.65%) and a high ROC-AUC (93.32%). The combination of Tomek+ENN resulted in maintained high precision (91.95%) and recall (81.63%), contributing to a balanced F1 score (86.48%) and a robust ROC-AUC (90.81%).

In summary, the resampling techniques induced trade-offs in the precision and recall of the CatBoost model. Tomek Links and Tomek+ENN demonstrated effectiveness in both precision and recall. SMOTE and SMOTE+ENN were effective in enhancing recall, but at the cost of precision. ENN offers a balanced approach, and the choice of resampling technique depends on the specific objectives and desired trade-offs between precision and recall.

- Precision analysis

Precision, which measures the accuracy of positive predictions, is crucial in cyber fraud detection to minimize false positives. Before resampling, CatBoost demonstrated exceptional precision of 98.75%, indicating a robust ability to accurately identify fraudulent transactions. However, after resampling, Tomek resampling was superior to RF, achieving a precision of 97.47%. This demonstrates the effectiveness of Tomek resampling in reducing

false positives while maintaining a high precision. In contrast, SMOTE and SMOTE + ENN showed a decrease in precision for several models, notably DT, indicating a potential trade-off between precision and the other metrics.

- Recall analysis

Recall that emphasizing the identification of true-positive cases is crucial in identifying fraudulent transactions. Before resampling, DT exhibited the highest recall of 81.63%. After resampling, SMOTE consistently increased the recall across various models, with RF achieving a recall of 0.8571. However, this increase in recall was accompanied by a decrease in the precision for certain models, highlighting the potential sensitivity to false positives.

- F1 score analysis

The F1 score, which balances the precision and recall, provides a comprehensive performance metric. Catboost exhibited the highest F1 score before resampling at 88.76%. After resampling, SMOTE showed an increase in F1 score for certain models, but it had a notable negative impact on DT's F1 score of DT. Tomek and ENN consistently demonstrated balanced improvements in F1 score, demonstrating their effectiveness in maintaining a harmonious balance between precision and recall. For instance, ENN increased the F1 score for RF by 83.60%.

- Overall performance

Considering the overall performance, SMOTE and SMOTE + ENN may be suitable for scenarios that prioritize recall. However, the observed decrease in precision for some models should be carefully weighed against the increase in the recall. The Tomek and ENN techniques offer a more balanced approach, enhancing recall without severely impacting the precision.

- Summary

Based on a comprehensive analysis of the precision, recall, and F1 score, the choice of resampling technique should align with specific business goals and constraints, considering the trade-offs between false positives and false negatives. Each resampling method exhibits varying effects on different models, and the optimal choice should align with business goals and constraints. If the primary objective is to reduce false positives and improve precision without sacrificing recall, the Tomek and ENN resampling techniques have emerged as favourable options. For example, after Tomek resampling, RF achieved a precision of 97.47%, showing a substantial improvement from the initial precision of 96.25%. The Recall, although not dramatically increased, remained reasonable at 78.57%.

However, if maximizing recall is critical, even at the expense of precision, SMOTE and SMOTE + ENN can be considered. SMOTE consistently increased the recall across models; however, it often led to a decrease in precision. SMOTE + ENN attempts to strike a balance between oversampling and undersampling, thereby providing a mixed impact on precision and recall. Ultimately, the choice should be driven by a thorough understanding of a business's tolerance for false positives and false negatives. If maintaining a high level of precision is crucial for minimizing the effects of false alarms, techniques such as Tomek or ENN, which exhibit improvements in precision across various models, may be more suitable. However, if the focus is on capturing as many fraudulent transactions as possible, even with a potential increase in false positives, SMOTE or SMOTE + ENN can be considered.

5. Conclusions

The development of a machine learning-based model for credit card cyber fraud detection is crucial for strengthening the security of financial transactions. In response

to increasingly sophisticated fraud methodologies, this study proposed and evaluated a comprehensive ML-based cyber fraud detection framework that emphasizes the balance between minimizing false positives and false negatives. Misclassification of fraudulent transactions can lead to severe financial and operational consequences for institutions, making this balance particularly critical. Seven classical machine learning classifiers—RF, DT, LR, KNN, SVM, XGBoost, and CatBoost—were evaluated both without resampling and in combination with multiple resampling strategies (Tomek, SMOTE, ENN, SMOTE+ENN, and Tomek+ENN) within a leakage-safe evaluation protocol to ensure reliability and reproducibility.

The results demonstrate that before applying resampling techniques, RF, DT, XGBoost, and CatBoost achieved strong overall performance, exhibiting a favorable balance between precision and recall under extreme class imbalance. LR showed limitations in balancing precision and recall, while KNN remained competitive due to its simplicity and effectiveness. SVM achieved high precision but suffered from reduced recall. Among all evaluated models, CatBoost consistently emerged as the most robust classifier, achieving superior precision–recall balance and effectively minimizing false positives while capturing fraudulent transactions. DT and RF also demonstrated reliable performance, with RF showing particularly high precision but leaving room for recall improvement. These findings provide practical guidance for selecting appropriate classifiers in credit card fraud detection systems.

The analysis further highlights the importance of aligning resampling strategies with specific business objectives and tolerance levels for false positives and false negatives. Tomek and ENN were found to be effective when precision is prioritized, while SMOTE and SMOTE+ENN improved recall at the cost of increased false positives. These trade-offs emphasize the need for context-aware model and resampling selection in operational fraud detection environments.

Although the dataset includes a temporal attribute, fully time-ordered validation (training on earlier transactions and testing on later ones) is not feasible due to the anonymized nature of the data and the absence of real transaction timestamps. Consequently, the reported results reflect static evaluation settings rather than deployment-time temporal behavior. Future work will explore temporal validation using datasets that provide explicit timestamp information.

Although deep learning approaches have shown promise in credit card fraud detection, their experimental evaluation is outside the scope of this study. The focus of this work is on classical machine learning models, which remain widely adopted in financial institutions due to their lower computational cost, reduced training complexity, and greater interpretability. Furthermore, the anonymized and PCA-transformed nature of the dataset limits the advantages typically associated with deep learning architectures. Future research will extend the proposed benchmarking framework to include deep learning baselines for comparative analysis.

Finally, this study acknowledges certain limitations, including the use of a single dataset and a restricted set of resampling techniques. Future work may explore additional resampling methods such as random oversampling, random undersampling, Adaptive Synthetic Sampling, and NearMiss, as well as evaluate the proposed framework across multiple datasets to further assess its generalizability.

Author Contributions: Conceptualisation, E.B. and X.Z.; methodology, E.B., X.Z. and T.K.; software, E.B., T.K. and H.T.; validation, E.B., X.Z. and H.T.; formal analysis, E.B. and T.K.; investigation, E.B., X.Z. and H.T.; resources, E.B. and T.K.; data curation, E.B. and X.Z.; writing—original draft preparation, E.B. and T.K.; writing—review and editing, E.B., H.T. and T.K.; visualisation, E.B. and

H.T.; supervision, E.B., X.Z. and H.T. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The dataset is available at <https://www.kaggle.com/mlg-ulb/creditcardfraud> (accessed on 1 January 2025). The dataset consisted of client transactions at a European bank in 2013. The real-world dataset consists of 284,807 credit card transactions.

Conflicts of Interest: The authors declare no conflicts of interest.

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